

MATH-3108-103-71037-2022FS-Linear Algebra I Exam 2

Noah Cain

TOTAL POINTS

94 / 100

QUESTION 1

1 Question 1 4 / 4

✓ - 0 pts Fully correct

QUESTION 2

2 Question 2 3 / 3

✓ - 0 pts Correct

QUESTION 3

3 Question 3 3 / 3

✓ - 0 pts Correct

QUESTION 4

4 Question 4 3 / 3

✓ - 0 pts Correct

QUESTION 5

5 Question 5 3 / 3

✓ - 0 pts Correct

QUESTION 6

6 Question 6 2 / 3

✓ - 1 pts One correct answer, one incorrect answer.
The correct answer is b.

QUESTION 7

7 Question 7 0 / 3

✓ - 3 pts Incorrect. The correct answer is c.

QUESTION 8

8 Question 8 3 / 3

✓ - 0 pts Correct

QUESTION 9

9 Question 9 3 / 3

✓ - 0 pts Correct

QUESTION 10

Question 10 20 pts

10.1 Question 10a 4 / 4

✓ - 0 pts Correct

10.2 Question 10b 8 / 8

✓ - 0 pts Correct

10.3 Question 10c 4 / 4

✓ - 0 pts Correct

10.4 Question 10d - first blank 2 / 2

✓ - 0 pts Correct

10.5 Question 10d - second blank 2 / 2

✓ - 0 pts Correct

QUESTION 11

11 Question 11 10 / 10

✓ - 0 pts Skipping/Not enough/Correct work shown

to initiate Cramer's Rule:

$$z = |A_3| / |A|$$

QUESTION 12

12 Question 12 10 / 10

✓ - 0 pts Correct

QUESTION 13

13 Question 13 6 / 8

✓ - 2 pts Final answer should be simplified

****completely **and **correctly**.**

![[Untitled.jpg]](/files/4fae1509-2a20-4f9d-97e6-1a61c99d4c9a)

QUESTION 14

14 Question 14 12 / 12

✓ - 0 pts Correct

QUESTION 15

15 Question 15 12 / 12

✓ - 0 pts Correct

Name: Noah Cain

Math 3108 – Linear Algebra
Exam 2
Prof. Runnion
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Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned completely off (silence or on vibrate is *not* acceptable) and put away for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) You must write **darkly** and **legibly** – this exam will be scanned for electronic grading.
- 4) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 5) Failure to follow directions specific to a problem will result in the loss of points.
- 6) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 7) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 8) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 9) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 10) Extra space has been provided on all problems. There should be more than sufficient space to work all problems in the space indicated. Thus, NO WORK APPEARING ON THE BACK OF PAGES WILL BE GRADED!
- 11) All work must appear within this exam packet to be graded. **No scratch paper is allowed under any circumstances.**
- 12) This packet has eight sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 13) Once this exam begins, you will have 50 minutes to complete your solutions.

DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO

Name: Noah Cain

Multiple Choice and True/False. Completely fill in the box corresponding to the correct answer(s).

1. (4 points) Let A be a 4×4 matrix. If the fourth row of A is the sum of the second and third rows, which of the following is true? (Select all that apply.)

☐ i) The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

☒ ii) The column space of A is a subspace of $\mathbb{R}^4$.

☐ iii) The rows of A span $\mathbb{R}^4$.

☒ iv) The determinant of A is zero.

2. (3 points) Let A be an $m \times n$ matrix. Which of the following is a subspace of $\mathbb{R}^n$?

☐ i) $\text{Col } A$ (the column space of A)

☒ ii) $\text{Nul } A$ (the null space of A)

☐ iii) Both $\text{Col } A$ and $\text{Nul } A$.

☐ iv) Neither $\text{Col } A$ nor $\text{Nul } A$.

3. (3 points) A basis for a vector space V is a _____ of vectors in V .

☐ a) minimal spanning set

☐ b) maximal linearly independent set

☒ c) both (a) and (b)

☐ d) neither (a) nor (b)

4. (3 points) Let A be a 4×6 matrix. The pivot columns of A are _____ linearly independent.

☒ a) always

☐ b) sometimes

☐ c) never

5. (3 points) If A is a 4×6 matrix, what is the smallest possible dimension of $\text{Nul } A$?

☐ 0

☐ 1

☒ 2

☐ 3

☐ 4

☐ 5

☐ 6

Name: Noah Cain

6. (3 points) Let $\{v_1, v_2, v_3, v_4, v_5\}$ be a linearly independent set of nonzero vectors from a vector space V . Which of the following statements are true? (Select all that apply.)

☒ a) $\text{Span}\{v_1, v_2, v_3, v_4, v_5\}$ is a subspace of $\mathbb{R}^5$.

☒ b) $\text{Span}\{v_1, v_2, v_3, v_4, v_5\}$ is a subspace of V .

☐ c) $\{v_1, v_2, v_3, v_4, v_5\}$ is a basis for $\mathbb{R}^5$.

☐ d) $\{v_1, v_2, v_3, v_4, v_5\}$ is a basis for V .

7. (3 points) Which of the following is a two-dimensional subspace of $\mathbb{R}^3$? (Select all that apply.)

☒ a) A line through the origin in $\mathbb{R}^3$.

☐ b) A line not through the origin in $\mathbb{R}^3$.

☐ c) A plane through the origin in $\mathbb{R}^3$.

☐ d) A plane not through the origin in $\mathbb{R}^3$.

☒ e) $\mathbb{R}^2$

8. (3 points) Given $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & j \end{vmatrix} = 7$, calculate $\begin{vmatrix} a & b & c \\ g & h & j \\ d & e & f \end{vmatrix} =$ _____.

☐ a) 7 ☒ b) -7 ☐ c) $\frac{1}{7}$ ☐ d) None of these

9. (3 points) Find the dimension of the subspace H of $\mathbb{R}^2$ spanned by

$$\begin{bmatrix} 1 \\ -5 \end{bmatrix}, \begin{bmatrix} -2 \\ 10 \end{bmatrix}, \text{ and } \begin{bmatrix} -3 \\ 15 \end{bmatrix}.$$

☐ a) 0 ☒ b) 1 ☐ c) 2 ☐ d) 3

$$\begin{bmatrix} 1 & -2 & -3 \\ -5 & 10 & 15 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

Name: Noah Cain

Open Answer. Show ALL your work!

10. The matrices A and B given below are row equivalent. (You do *not* need to check this.)

$$A = \begin{bmatrix} 2 & -4 & -1 & -3 & 5 & 2 \\ -1 & 2 & -4 & -3 & -7 & -1 \\ -1 & 2 & -13 & -12 & -16 & -1 \\ 5 & -10 & 5 & -10 & 20 & 5 \end{bmatrix}$$

$$B = \begin{matrix} & x_2 & x_3 & x_4 & x_5 & x_6 \\ \begin{bmatrix} 1 & -2 & 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} & \begin{matrix} 0 \\ 0 \\ 0 \\ 0 \end{matrix} \end{matrix}$$

- a) (4 points) Find a basis for the column space of A .

$$\text{Basis for } \text{Col } A = \text{span} \left\{ \begin{bmatrix} 2 \\ -1 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} -1 \\ -4 \\ -13 \\ 5 \end{bmatrix}, \begin{bmatrix} -3 \\ -3 \\ -12 \\ -10 \end{bmatrix} \right\}$$

- b) (8 points) Find a basis for the null space of A .

$$\begin{aligned} \checkmark \quad x_1 &= 2x_2 - 3x_5 - 1x_6 \\ \checkmark \quad x_2 &= x_2 \equiv \text{free} \\ \checkmark \quad x_3 &= -x_5 \\ \checkmark \quad x_4 &= 0 \\ \checkmark \quad x_5 &= x_5 \equiv \text{free} \\ \checkmark \quad x_6 &= x_6 \equiv \text{free} \end{aligned}$$

$$\vec{x} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_6 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Basis null } A = \text{span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Name: Noah Cain

9. (continued) The matrices A and B given below are still row equivalent.

$$A = \begin{bmatrix} 2 & -4 & -1 & -3 & 5 & 2 \\ -1 & 2 & -4 & -3 & -7 & -1 \\ -1 & 2 & -13 & -12 & -16 & -1 \\ 5 & -10 & 5 & -10 & 20 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & -2 & 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- c) (4 points) Find a basis for the row space of A .

Basis row $A = \text{span} \left\{ \begin{bmatrix} 2 \\ -4 \\ -1 \\ -3 \\ 5 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ -4 \\ -3 \\ -7 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ -13 \\ -12 \\ -16 \\ -1 \end{bmatrix} \right\}$

- d) (2 points each) Find each requested quantity.

rank $A = \boxed{3}$

nullity $A = \boxed{3}$

11. (10 points) Consider the system of equations
$$\begin{cases} 2x - 3y + 4z = 1 \\ x + 6z = 0 \\ 3x - 2y = 5 \end{cases}$$

Given that $\begin{vmatrix} 2 & -3 & 4 \\ 1 & 0 & 6 \\ 3 & -2 & 0 \end{vmatrix} = -38$, use Cramer's Rule to solve for the variable z only.

$$D_2 = \begin{vmatrix} 2 & -3 & 1 \\ 1 & 0 & 0 \\ 3 & -2 & 5 \end{vmatrix} = 2(0-0) + 3(5-0) + 1(-2-0) = 0 + 15 - 2 = 13$$

$$z = \frac{13}{-38} = \frac{D_2}{|A|} ?$$

Name: Noah Cain

12. (10 points) Show that $H = \left\{ \begin{bmatrix} b-c \\ 2b+c+d \\ 5c-4d \\ d \end{bmatrix} : b, c, d \text{ real} \right\}$ is a subspace of $\mathbb{R}^4$ by finding a

matrix A for which $H = \text{Col } A$.

$$H = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 1 & 1 \\ 0 & 5 & -4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} b \\ c \\ d \end{bmatrix} \quad A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 1 & 1 \\ 0 & 5 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$\text{Col } A \therefore \text{subspace of } \mathbb{R}^4$

so, H is a subspace of $\mathbb{R}^4$

13. (8 points) Assume that the matrices shown below are partitioned conformably for block multiplication. Compute the product.

$$\begin{bmatrix} I & 0 \\ -E & I \end{bmatrix} \begin{bmatrix} W & X \\ Y & Z \end{bmatrix}$$

Block product : $\begin{bmatrix} (IW + 0Y) & (IX + 0Z) \\ (-EW + IY) & (-EX + IZ) \end{bmatrix}$

Name: Noah Cain

14. (12 points) Compute the LU factorization of $A = \begin{bmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{bmatrix}$.

$$\begin{bmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 10 & 4 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

$$U = \begin{bmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{bmatrix}$$

Name: Noah Cain

15. (12 points) Determine whether the set of all polynomials of the form $p(t) = a + t^2$ where a and b are real is a subspace of $\mathbb{P}_2$. If it is, you must fully justify that it is a subspace. If it is not, you must fully justify that it fails to meet one of the necessary conditions for being a subspace.

to find if it is subspace

①

$\vec{0}$ must be contained!

because " t^2 " contains no leading variable

the zero vector is not contained in the set

So $p(t)$ cannot be a subspace of
 $\mathbb{P}_2$ or $a + bt + ct^2$.